

Boolean Algebra: Proving Basic Identities

This document presents rigorous proofs for fundamental Boolean algebra identities, including Idempotent Laws, Annulment Laws, Absorption Laws, and De Morgan's Law[1][2].

1. Idempotent Law (OR): $x + x = x$

Proof:

- $x + x = (x \cdot 1) + (x \cdot 1)$ \hfill (Identity Law: $x = x \cdot 1$)
- $x + x = x(1 + 1)$ \hfill (Distributive Law)
- Since $1 + 1 = 1$ in Boolean algebra: \hfill (Property of Boolean 1)
- $x + x = x(1)$ \hfill (Substitution)
- $x + x = x$ \hfill (Identity Law: $x \cdot 1 = x$)

Conclusion: $x + x = x$ (Idempotent Law for OR)[1][3]

2. Idempotent Law (AND): $x \cdot x = x$

Proof:

- $x \cdot x = x \cdot x + 0$ \hfill (Identity Law: $x + 0 = x$)
- Substitute $0 = x \cdot x'$: \hfill (Complement Law)
- $x \cdot x = x \cdot x + x \cdot x'$ \hfill (Substitution)
- Factor out x : \hfill (Distributive Law)
- $x \cdot x = x(x + x')$ \hfill (Factorization)
- Since $x + x' = 1$: \hfill (Complement Law)
- $x \cdot x = x(1)$ \hfill (Substitution)
- $x \cdot x = x$ \hfill (Identity Law: $x \cdot 1 = x$)

Conclusion: $x \cdot x = x$ (Idempotent Law for AND)[1][3]

3. Annulment Law (OR): $x + 1 = 1$

Proof:

- $x + 1 = 1 \cdot (x + 1)$ \hfill (Identity Law: $1 \cdot x = x$)
- Substitute $1 = x + x'$: \hfill (Complement Law)
- $x + 1 = (x + x')(x + 1)$ \hfill (Substitution)
- Use Distributive Law $(a + b)(a + c) = a + bc$: \hfill (Distribution)
- $x + 1 = x + (x' \cdot 1)$ \hfill (Simplification)
- $x + 1 = x + x'$ \hfill (Identity Law: $x' \cdot 1 = x'$)
- Since $x + x' = 1$: \hfill (Complement Law)
- $x + 1 = 1$ \hfill (Substitution)

Conclusion: $x + 1 = 1$ (Annulment Law for OR)[2][3]

4. Annulment Law (AND): $x \cdot 0 = 0$

Proof:

1. $x \cdot 0 = x \cdot 0 + 0$ \hfill (Identity Law: $x + 0 = x$)
2. Substitute $0 = x \cdot x'$: \hfill (Complement Law)
3. $x \cdot 0 = x \cdot 0 + x \cdot x'$ \hfill (Substitution)
4. Factor out x : \hfill (Distributive Law)
5. $x \cdot 0 = x(0 + x')$ \hfill (Factorization)
6. $x \cdot 0 = x(x')$ \hfill (Identity: $0 + x' = x'$)
7. Since $x \cdot x' = 0$: \hfill (Complement Law)
8. $x \cdot 0 = 0$ \hfill (Substitution)

Conclusion: $x \cdot 0 = 0$ (Annulment Law for AND)[2][3]

5. Absorption Law (Form 1): $x + xy = x$

Proof:

1. $x + xy = x \cdot 1 + xy$ \hfill (Identity Law: $x = x \cdot 1$)
2. Factor out x : \hfill (Distributive Law)
3. $x + xy = x(1 + y)$ \hfill (Factorization)
4. Using the property $1 + y = 1$: \hfill (Annulment Law for OR)
5. $x + xy = x(1)$ \hfill (Substitution)
6. $x + xy = x$ \hfill (Identity Law: $x \cdot 1 = x$)

Conclusion: $x + xy = x$ (Absorption Law)[1][4]

6. Absorption Law (Form 2): $x(x + y) = x$

Proof:

1. Expand using the Distributive Law: \hfill (Distributive Property)
2. $x(x + y) = x \cdot x + x \cdot y$ \hfill (Distribution)
3. Using the property $x \cdot x = x$ from Idempotent Law: \hfill (Idempotent Law for AND)
4. $x(x + y) = x + xy$ \hfill (Substitution)
5. Factor out x : \hfill (Distributive Law)
6. $x(x + y) = x(1 + y)$ \hfill (Factorization)
7. Since $1 + y = 1$: \hfill (Annulment Law for OR)
8. $x(x + y) = x(1)$ \hfill (Substitution)
9. $x(x + y) = x$ \hfill (Identity Law: $x \cdot 1 = x$)

Conclusion: $x(x + y) = x$ (Absorption Law)[1][4]

7. De Morgan's Law: $(x + y)' = x'y'$

To prove that $x'y'$ is the true complement of $(x + y)$, we must demonstrate two fundamental conditions that define a Boolean complement[5][6]:

- **Condition A:** Their sum equals 1 $\rightarrow (x + y) + (x'y') = 1$
- **Condition B:** Their product equals 0 $\rightarrow (x + y)(x'y') = 0$

Step 1: Verify the Sum (Condition A)

Goal: Show that $(x + y) + x'y' = 1$

1. $(x + y) + x'y'$ \hfill (Original expression)
2. Use Distributive Law $(A + BC) = (A + B)(A + C)$: \hfill (Distribution)
3. $= (x + y + x')(x + y + y')$ \hfill (Expansion)
4. Rearrange terms using $x + x' = 1$ and $y + y' = 1$: \hfill (Complement Law)
5. $= (1 + y)(x + 1)$ \hfill (Simplification)
6. Since $1 + \text{anything} = 1$: \hfill (Annulment Law)
7. $= 1 \cdot 1$ \hfill (Substitution)
8. $= 1$ \hfill (**Condition A Met**)

Step 2: Verify the Product (Condition B)

Goal: Show that $(x + y)(x'y') = 0$

1. $(x + y)(x'y')$ \hfill (Original expression)
2. Distribute $x'y'$ to both terms: \hfill (Distributive Law)
3. $= x(x'y') + y(x'y')$ \hfill (Distribution)
4. $= (x \cdot x')y' + x'(y \cdot y')$ \hfill (Associativity)
5. Since $x \cdot x' = 0$ and $y \cdot y' = 0$: \hfill (Complement Law)
6. $= (0)y' + x'(0)$ \hfill (Substitution)
7. $= 0 + 0$ \hfill (Annulment Law for AND)
8. $= 0$ \hfill (**Condition B Met**)

Conclusion

Since both conditions are satisfied:

- The sum $(x + y) + (x'y') = 1$
- The product $(x + y)(x'y') = 0$

Therefore, $x'y'$ is the Boolean complement of $(x + y)$, proving De Morgan's Law[5][6]:

$$\boxed{(x + y)' = x' \cdot y'}$$

This law is fundamental in digital logic design and circuit simplification, enabling the transformation of OR gates with inverted outputs into AND gates with inverted inputs[7].

Summary of Proven Identities

Law	Identity	Type
Idempotent (OR)	$x + x = x$	Basic Identity
Idempotent (AND)	$x \cdot x = x$	Basic Identity
Annulment (OR)	$x + 1 = 1$	Basic Identity
Annulment (AND)	$x \cdot 0 = 0$	Basic Identity
Absorption (Form 1)	$x + xy = x$	Simplification
Absorption (Form 2)	$x(x + y) = x$	Simplification
De Morgan's Law	$(x + y)' = x'y'$	Transformation

Table 1: Summary of Boolean algebra identities proven in this document

These identities form the foundation for Boolean algebra manipulation and are extensively used in digital circuit design, computer science, and logic optimization[1][2][3].

References

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